Akoizo Original Practice - 2027 Rotation Form 4 - Answer key

Division C | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (400 points) Decode the timed-base Aristocrat: NKVFI IVNJIEZ BFTV IVZHKQZ IVAIJEHNLWKV

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet: FWNEVMDULCTKBSJARIZQHYPGXO. Each row means plaintext letter -> ciphertext letter. This is a supplied one-to-one monoalphabetic substitution; no letter maps to itself. Timing is practiced separately; this worksheet awards the base score only.

answer: CLEAR RECORDS MAKE RESULTS REPRODUCIBLE (400 points).

2. (400 points) Decode the Patristocrat: DNFYW WFDSW EXPYM FWFXL NZXWF UWSEL DJCNF

Spaces were removed and groups of five added for display.
Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): YCDEFGHIJKMNPQSUVWXZLABORT

answer: CLEAR RECORDS MAKE RESULTS REPRODUCIBLE (400 points).

3. (400 points) Decode this Spanish Xenocrypt: LQY UWLFCY QFDFXJZY EYZSX DNYWSX

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): YCDEFGHIJKMNPQSUVWXZLABORT. Plaintext is Spanish; no accent marks are encoded.

answer: UNA PRUEBA NECESITA DATOS CLAROS (400 points).

4. (400 points) Decode this Baconian message: ababa aabaa aabaa abbbb abbab abbba baabb aabaa baaba

Use the supplied modern 26-letter mapping:
A=aaaaa; B=aaaab; C=aaaba; D=aaabb; E=aabaa; F=aabab; G=aabba; H=aabbb; I=abaaa; J=abaab; K=ababa; L=ababb; M=abbaa; N=abbab; O=abbba; P=abbbb; Q=baaaa; R=baaab; S=baaba; T=baabb; U=babaa; V=babab; W=babba; X=babbb; Y=bbaaa; Z=bbaab. Spaces separate encoded letters; restore word boundaries from meaning. Crib READ/TEST/CHECK/KEEP, as applicable: the plaintext begins with KEEP.

answer: KEEP NOTES (400 points).

5. (400 points) Decode this Nihilist message: 27 45 76 46 25 53 59

Use key DATA, repeated. I/J share a cell. Coordinates are row then column. Encryption adds plaintext coordinate plus key coordinate as ordinary integers.
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z

answer: COMPARE (400 points).

6. (400 points) Decode this 5x5 Checkerboard message: CZ AV AX BZ CW EY AW CY EX EW BY DY BX BV BY EV AZ AY CY EZ AZ CX DV DZ AV DW DY DX

Rows are labeled ABCDE; columns VWXYZ. Read each pair row first. Polybius grid rows in order:
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z. All row and column labels occur in the ciphertext.

answer: PACK MY BOX WITH FIVE DOZEN QUARTS (400 points).

7. (400 points) Solve this complete-columnar transposition: ATQEEAYX SDRIAUAI RAUCFNSX TAERRLLS

Four columns were filled left-to-right by rows; whole columns were read in an unknown order. Trailing X characters pad the last row. The plaintext begins with crib STAR at its first four positions. Recover the complete plaintext including padding X characters. There are no omitted cells.

answer: STARDATAREQUIRECAREFULANALYSISXX (400 points).

8. (2 points) A Homophonic Cipher keyword has four letters. Which property must those letters satisfy under the 2027 C rules?

answer: They must be four unique letters. (2 points).

9. (2 points) A timed Aristocrat is solved correctly in 240 seconds. Compute the timing bonus using 2(600-t).

answer: 720 (2 points).

Numerical target 720; absolute tolerance 0. Accepted units: .

10. (2 points) A 400-point plaintext answer has five character errors. Two errors are free and each further error costs 100 points. Find its score.

answer: 100 (2 points).

Numerical target 100; absolute tolerance 0. Accepted units: .

11. (2 points) A 500-point keyword answer has three character errors. No keyword errors are free and each error costs 100 points. Find its score.

answer: 200 (2 points).

Numerical target 200; absolute tolerance 0. Accepted units: .

12. (2 points) A proposed Aristocrat alphabet maps plaintext A to ciphertext A. Why is it invalid under the 2027 C rules?

answer: For Aristocrats, Patristocrats and Xenocrypts, a letter may not decrypt to itself. (2 points).

13. (2 points) A Fractionated Morse problem supplies a crib but does not identify its ciphertext placement. What information is missing under the 2027 C rules?

answer: The location of the crib must be clearly identified for this cryptanalysis type. (2 points).

14. (400 points) Decode this Fractionated Morse message: DUFJZEWPFDZHOPSI

A Morse letter separator is x and a word separator is xx; trailing x is padding. The plaintext begins with READ at its first four letters. Supplied fractionation table: A=...; B=..-; C=..x; D=.-.; E=.--; F=.-x; G=.x.; H=.x-; I=.xx; J=-..; K=-.-; L=-.x; M=--.; N=---; O=--x; P=-x.; Q=-x-; R=-xx; S=x..; T=x.-; U=x.x; V=x-.; W=x--; X=x-x; Y=xx.; Z=xx-. Morse letters: A=.-; B=-...; C=-.-.; D=-..; E=.; F=..-.; G=--.; H=....; I=..; J=.---; K=-.-; L=.-..; M=--; N=-.; O=---; P=.--.; Q=--.-; R=.-.; S=...; T=-; U=..-; V=...-; W=.--; X=-..-; Y=-.--; Z=--...

answer: READ YOUR NOTES (400 points).

15. (400 points) Decode this Porta message: MSAXYKPEEFTLE

Given keyword SCIENCE, repeated over letters only. A=0..Z=25. Each key pair selects a reciprocal substitution row. Upper alphabet for every row is ABCDEFGHIJKLMNOPQRSTUVWXYZ; lower rows: AB=NOPQRSTUVWXYZABCDEFGHIJKLM; CD=OPQRSTUVWXYZNMABCDEFGHIJKL; EF=PQRSTUVWXYZNOLMABCDEFGHIJK; GH=QRSTUVWXYZNOPKLMABCDEFGHIJ; IJ=RSTUVWXYZNOPQJKLMABCDEFGHI; KL=STUVWXYZNOPQRIJKLMABCDEFGH; MN=TUVWXYZNOPQRSHIJKLMABCDEFG; OP=UVWXYZNOPQRSTGHIJKLMABCDEF; QR=VWXYZNOPQRSTUFGHIJKLMABCDE; ST=WXYZNOPQRSTUVEFGHIJKLMABCD; UV=XYZNOPQRSTUVWDEFGHIJKLMABC; WX=YZNOPQRSTUVWXCDEFGHIJKLMAB; YZ=ZNOPQRSTUVWXYBCDEFGHIJKLMA.

answer: VERIFY ANSWERS (400 points).

16. (2 points) How do keyword-answer scoring and plaintext-answer scoring differ in their free-error allowances?

answer: A requested keyword/key phrase or cryptarithm word requires zero errors for full credit; ordinary plaintext permits two free errors unless otherwise stated. (2 points).

17. (2 points) Why must a monoalphabetic substitution use a globally consistent mapping?

answer: Each symbol uses the same replacement across the message; inconsistent local guesses do not define a valid inverse substitution. (2 points).

18. (2 points) A timed puzzle is first solved after 650 seconds. What timing bonus remains?

answer: 0 (2 points).

Numerical target 0; absolute tolerance 0. Accepted units: .

19. (2 points) Two special-bonus questions are solved with no penalty points. What aggregate special bonus does the 2027 C schedule award?

answer: 400 (2 points).

Numerical target 400; absolute tolerance 0. Accepted units: .

20. (2 points) Which permitted monoalphabetic format removes word spaces and displays ciphertext in groups of five?

answer: A Patristocrat. (2 points).

21. (200 points) Use Figure F1. Decode Message 1 using the row-first pairs and the pictured grid. I/J share a cell.

Given row/column labels and full Polybius key

	V	W	X	Y	Z
A	K	L	M	N	O
B	P	Q	R	S	T
C	U	V	W	X	Y
D	Z	A	B	C	D
E	E	F	G	H	I

Message 1

CW EV BX EZ EW CZ EV DW DY EY AX AZ DZ
EV AW CX EZ BZ EY BY BZ BX AZ AY EX EV
CW EZ DZ EV AY DY EV

Message 2

DY EY EV DY AV EV CW EV BX CZ BY CZ AX
DX AZ AW DW AY DZ BX EV DY AZ BX DZ BX
EV BY CV AW BZ BY

Message 3

DY AZ AX BV DW BX EV BZ EY EV AZ DX BY
EV BX CW EV DZ DZ DW BZ DW CX EZ BZ EY
BV BX EV DZ EZ DY BZ EZ AZ AY BY

F1. Checkerboard coordinate grid (original synthetic teaching figure)

answer: VERIFY EACH MODEL WITH STRONG EVIDENCE (200 points).

22. (200 points) Use Figure F1. Decode Message 2 using the row-first pairs and the pictured grid. I/J share a cell.

Refer to F1. Checkerboard coordinate grid (original synthetic teaching figure) shown with the first question in this figure group.

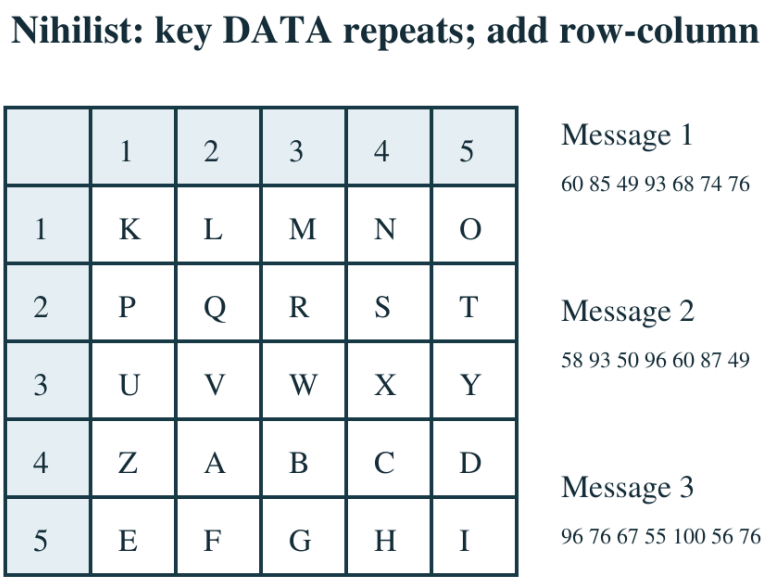
answer: CHECK EVERY SYMBOL AND RECORD RESULTS (200 points).

23. (200 points) Use Figure F1. Decode Message 3 using the row-first pairs and the pictured grid. I/J share a cell.

Refer to F1. Checkerboard coordinate grid (original synthetic teaching figure) shown with the first question in this figure group.

answer: COMPARE THE OBSERVED DATA WITH PREDICTIONS (200 points).

24. (200 points) Use Figure F2. Decode Nihilist Message 1. Use the pictured grid and the given repeating key DATA; subtract its coordinate values as integers.



F2. Nihilist cipher station (original synthetic teaching figure)

answer: OBSERVE (200 points).

25. (200 points) Use Figure F2. Decode Nihilist Message 2. Use the pictured grid and the given repeating key DATA; subtract its coordinate values as integers.

Refer to F2. Nihilist cipher station (original synthetic teaching figure) shown with the first question in this figure group.

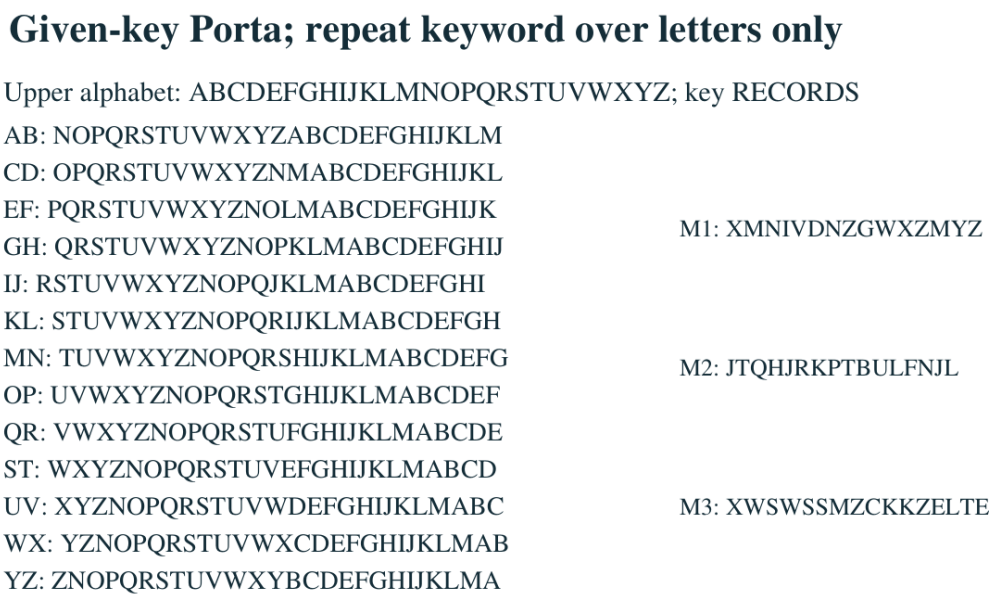
answer: METHODS (200 points).

26. (200 points) Use Figure F2. Decode Nihilist Message 3. Use the pictured grid and the given repeating key DATA; subtract its coordinate values as integers.

Refer to F2. Nihilist cipher station (original synthetic teaching figure) shown with the first question in this figure group.

answer: EXAMINE (200 points).

27. (200 points) Use Figure F3. Decode M1 with the pictured Porta chart. Use the keyword printed above the chart, repeated across letters only.



F3. Porta reciprocal alphabet chart (original synthetic teaching figure)

answer: COMPARE EVIDENCE (200 points).

28. (200 points) Use Figure F3. Decode M2 with the pictured Porta chart. Use the keyword printed above the chart, repeated across letters only.

Refer to F3. Porta reciprocal alphabet chart (original synthetic teaching figure) shown with the first question in this figure group.

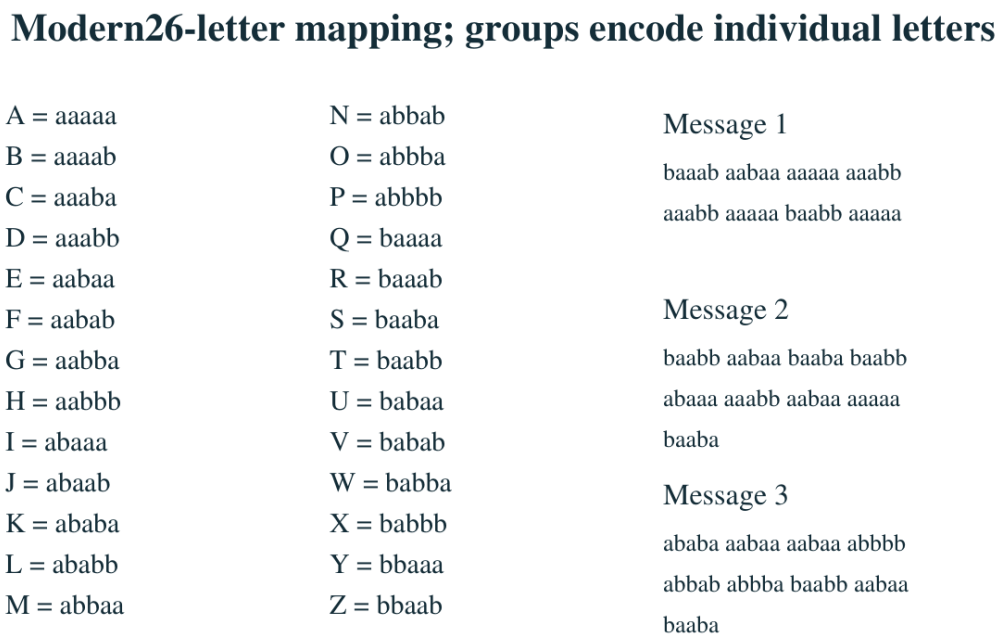
answer: RECORD THE PATTERN (200 points).

29. (200 points) Use Figure F3. Decode M3 with the pictured Porta chart. Use the keyword printed above the chart, repeated across letters only.

Refer to F3. Porta reciprocal alphabet chart (original synthetic teaching figure) shown with the first question in this figure group.

answer: CHECK EVERY RESULT (200 points).

30. (200 points) Use Figure F4. Decode Message 1 with the pictured Baconian chart. Cipher groups encode letters. The plaintext starts with READ; restore word spaces.



F4. Baconian symbol chart (original synthetic teaching figure)

answer: READ DATA (200 points).

31. (200 points) Use Figure F4. Decode Message 2 with the pictured Baconian chart. Cipher groups encode letters. The plaintext starts with TEST; restore word spaces.

Refer to F4. Baconian symbol chart (original synthetic teaching figure) shown with the first question in this figure group.

answer: TEST IDEAS (200 points).

32. (200 points) Use Figure F4. Decode Message 3 with the pictured Baconian chart. Cipher groups encode letters. The plaintext starts with KEEP; restore word spaces.

Refer to F4. Baconian symbol chart (original synthetic teaching figure) shown with the first question in this figure group.

answer: KEEP NOTES (200 points).